

Emulating Hyper-Arity Topology on von Neumann Architecture: The vHPU Engine

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Abstract

Traditional neural networks reliant on continuous floating-point matrix multiplications suffer from massive "Virtual Heat"—computational latency and energy waste—when simulating discrete topological states. In this paper, we introduce the Virtual Hyper-Arity Processing Unit (vHPU), a software emulator designed to map Poly-Algebraic logic directly onto von Neumann architectures. By replacing dense Multi-Layer Perceptrons (MLPs) with discrete Rulial Inversion instructions, we demonstrate an empirical 12.4x speedup (a 91.95% reduction in Virtual Heat) when advecting a 1D Burgers' equation shockwave. This lays the software foundation for hardware-accelerated Acoustic-Fluidic HPUs.

1 Introduction

Modern physical simulations utilizing AI (such as Graph Neural Networks for N-body problems) are severely bottlenecked by the underlying hardware. PyTorch and standard deep learning frameworks operate via dense matrix operations (GEMMs). When the underlying physical state is discrete (e.g., topological bonds breaking and forming), calculating continuous probability gradients across an overparameterized MLP is incredibly inefficient. We term this efficiency loss "Virtual Heat."

2 The vHPU Architecture

The vHPU bypasses continuous backpropagation by invoking specific Poly-Algebraic discrete instructions. Rather than learning a highly complex non-linear transformation via dense layers ($y = \text{GELU}(W_2(\text{GELU}(W_1x)))$) to simply shift a spatial array, the vHPU recognizes the topological invariant of the state and executes a discrete *Rulial Invert* operation.

2.1 Discrete Advection over Continuous Regression

In the case of a 1D propagating shockwave front (governed by a simplified Burgers' equation limit), the physical system maintains local amplitude but shifts spatially. The vHPU represents this state as a 1D lattice. A standard MLP requires $O(N^2)$ memory and operations per layer to regress the new state. The vHPU utilizes an $O(N)$ discrete pointer shift (Rulial Invert).

3 Empirical Benchmarks: The Virtual Heat Profile

We developed a highly simplified 1D Burgers' shockwave dataset ($N = 4096$ spatial resolution) and benchmarked the vHPU emulator against an optimized PyTorch MLP.

3.1 Results

Testing across batches of 1000 configurations for 50 iterations yielded conclusive evidence of the Virtual Heat bottleneck:

- **Standard PyTorch MLP:** 91.99 ms per forward pass.
- **vHPU Rulial Invert:** 7.40 ms per forward pass.
- **Speedup: 12.43x**
- **Virtual Heat Reduction: 91.95%**

The massive reduction in clock cycles proves that translating Poly-Algebraic graph routing directly to the hardware level (even on a von Neumann emulator) offers transformative efficiency gains for AI-driven physics engines.

4 Conclusion and Future Hardware

The software vHPU emulator successfully proves that enforcing discrete Poly-Algebraic logic over standard continuous regression eliminates the vast majority of computational waste. Phase 3 of our research will transpose this logic onto a physical Acoustic-Fluidic table, completely bypassing silicon limitations to achieve zero-standby thermodynamic simulation.